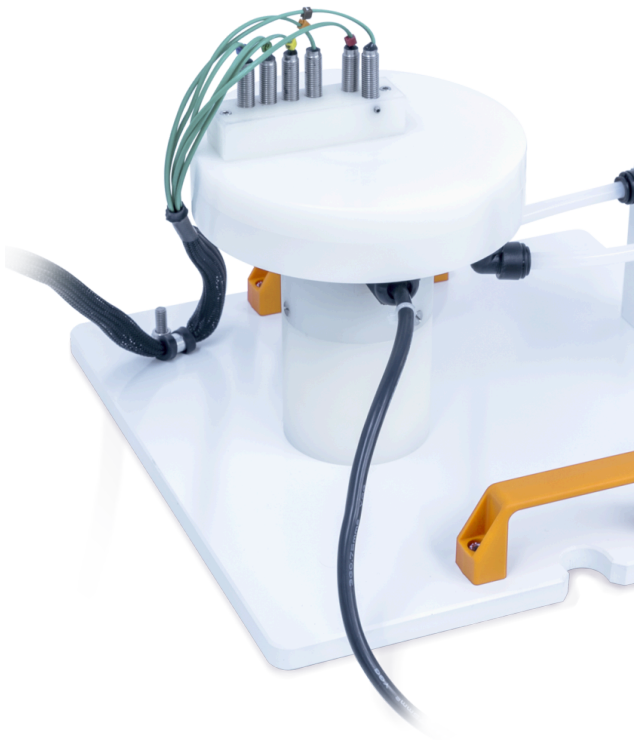




HT12X – Radial Heat Conduction

The arrangement, using a solid metal disk with temperature measurements at different radii and heat flow radially outward from the centre to the periphery, enables the temperature distribution and flow of heat by radial conduction to be investigated.

On the HT12X the heater power and the cooling water flow rate are regulated by PID controlled via the HT10X via the computer software.

PRODUCT CODE: HT12X

Categories: Educational, HT Series - Computer Controlled Heat Exchange / Heat Transfer, HT12X – Radial Heat Conduction

Tags: Computer-Controlled Radial Heat Conduction , Cooling Water Flow Rate , Flow Of Heat , Heat Flow , Heat Transfer And Thermodynamics , Heater Power , HT Series , HT12 , HT12C , K-Type Thermocouples , Pressure Regulator , Quick-Release Connections , Radial Conduction , Radial Heat Conduction , Steady-State Conduction , Temperature Distribution , Temperature Gradient , Temperature Measurements , Thermal Conductivity K , Turbine Type Flow Sensor

DESCRIPTION

The arrangement, using a solid metal disk with temperature measurements at different radii and heat flow radially outward from the centre to the periphery, enables the temperature distribution and flow of heat by radial conduction to be investigated.

On the HT12X the heater power and the cooling water flow rate are regulated by PID controlled via the HT10X via the computer software.

The accessory comprises a solid disk of material, which is heated at the centre and cooled at the periphery to create a radial temperature difference with corresponding radial flow of heat by conduction

- Six K-type thermocouples are positioned at different radii in the heated disk to indicate the temperature gradient from the central heated core to the periphery of the disk
- The radial distance between each thermocouple in the disk is 10mm
- Quick-release connections facilitate rapid connection of the cooling tube to a cold water supply
- Heater can be voltage or PID controlled allowing for the temperature set-point to be achieved rapidly and maintained within 0.1°C
- The HT10X service unit provides the accessory with the cooling water system:

- A pressure regulator is incorporated to minimise the effect of fluctuations in the supply pressure
- The cooling water flow rate is measured by a turbine type flow sensor
- A control valve on the HT10X service unit allows the flow of cooling water to be varied, if required, over the operating range of 0-1.5 l/min. Valve can be PID controlled for improved supply pressure disturbance reduction

TECHNICAL SPECIFICATIONS

A small-scale accessory to introduce students to the principles of radial heat conduction, and to allow the conductivity of a solid brass disk to be measured:

- A small-scale accessory to introduce students to the principles of radial heat conduction, and to allow the conductivity of a solid brass disk to be measured
- Comprised of a brass disk which with a PID controlled heating section heated at the centre and a cooling water tube attached to the periphery to create a radial temperature difference with corresponding radial flow of heat by conduction
- Conduction disk 110mm diameter and 3.2mm thick
- Heater power must be variable up to 100 Watts
- Six K-type thermocouples positioned at different radii in the heated disk to indicate the temperature gradient from the central heated core to the periphery of the disk.
- PID Controller Water flow rate variable up to 1.5 l/min

FEATURES & BENEFITS

- Quick release connectors fitted to all accessories that require water cooling (non drip)
- Software supplied as standard with exercise and calculations screens
- Manual includes exercises into understanding radial heat conduction
- Many experiments available

Experimental Capabilities

- Understanding the use of the Fourier rate equation in determining rate of heat flow through solid materials
- Measuring the temperature distribution for steady-state conduction of energy through the wall of a cylinder (radial energy flow)
- Determining the constant of proportionality (thermal conductivity k) of the disk material

Software

A full armBus educational software suite is provided with the HT10X for all the Armfield heat transfer accessories.

Individual experiment interfaces and displayed data types vary to match selected experiment accessories, common examples are listed below:

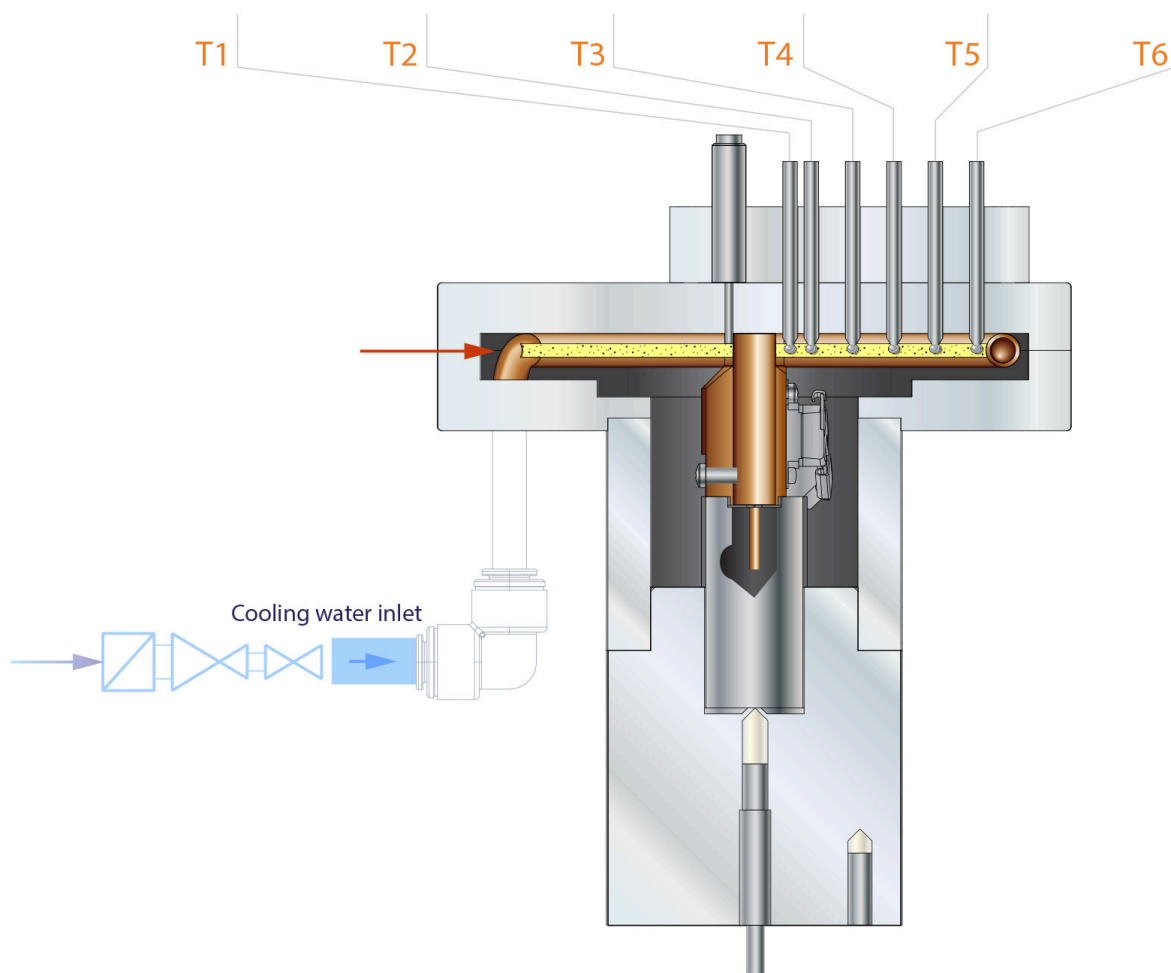
- Temperatures and other signals such as flow rates, heater voltage and current are displayed on a diagrammatic representation of the equipment
- Control outputs are operated by a control slider or typing in a value between 0 and 100%. Sensor values are read directly in engineering units
- PID automatic control option is available, permitting the temperature set-point to be achieved quickly and precisely, coping with disturbance rejection
- A maintenance screen allows for PID parameter adjustment and thermocouple calibration
- Data from the sensors are logged into a spreadsheet format, with operator control over the sampling intervals (or 'single-shot')
- Sophisticated graph-plotting facilities are provided, including plotting of both measured and calculated values
- Graphs update in real-time as the samples are being taken

- Processing of measured values to obtain calculated values is linked to the questions and answers to ensure student understanding
- Experiment data samples are saved, or exported in Microsoft Excel format
- Real-time sensor data is displayed independently from the data logging. It is possible to check the recent history graphical display to inspect the temperature stability prior to taking a sample



Armfield Radial Heat Conduction – HT12X Mimic Screen

HT12X Diagram



Armfield HT12X Diagram

Essential Accessories

HT10X Computer-Controlled Heat Transfer Service Unit

Requirements

All electrical requirements and Cold water supply are obtained from the **HT10X** service unit

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.05m³

Gross weight: 6kg

Dimensions



Length: 0.35m

Width: 0.18m

Height: 0.19m

Order Codes

HT12X

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